Production and Growth

Mbak 9512: Economics

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Review

- Consumer price index (CPI) and GDP deflator

$$\text{CPI} = \frac{\text{Price of basket of goods and services in current year}}{\text{Price of basket of goods and services in base year}}$$

$$\text{GDP deflator} = \frac{\text{Nominal GDP}}{\text{Real GDP}} \times 100$$

- Inflation

$$\pi_2 = \frac{P_2 - P_1}{P_1} \times 100$$

- Real vs. Nominal
  - purchasing power

Readings

- Chapter 7. Production and Growth
  - Economic Growth
  - Production
  - Economic Growth and Public Policy

Economic Growth around the World

- Real GDP per person
  - Living standard
  - Vary widely from country to country
- Growth rate
  - How rapidly real GDP per person grew in the typical year
- Because of differences in growth rates
  - Ranking of countries by income changes substantially over time

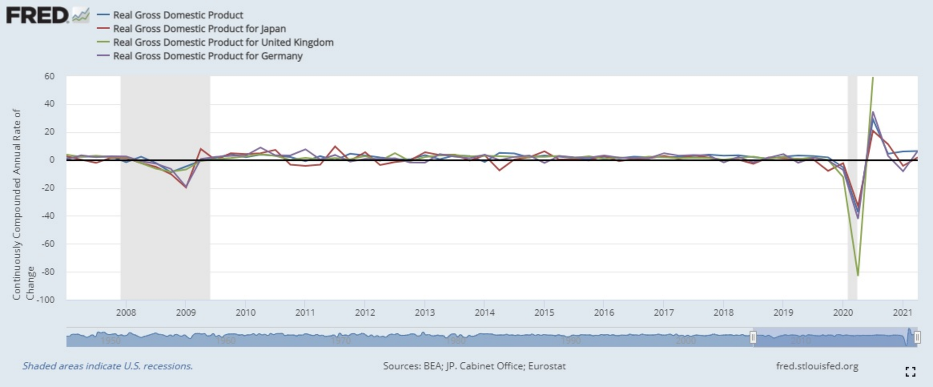
Economic Growth around the World

Country	Period	Real GDP per Person: At Beginning of Period[a]	Real GDP per Person: At End of Period[a]	Growth Rate (per year)
Brazil	1900–2014	$ 828	$15,590	2.61%
Japan	1890–2014	1,600	37,920	2.59
China	1900–2014	762	13,170	2.53
Mexico	1900–2014	1,233	16,640	2.31
Germany	1870–2014	2,324	46,850	2.11
Indonesia	1900–2014	948	10,190	2.10
Canada	1870–2014	2,527	43,360	1.99
India	1900–2014	718	5,630	1.82
United States	1870–2014	4,264	55,860	1.80
Pakistan	1900–2014	785	5,090	1.65
Argentina	1900–2014	2,440	12,510	1.44
Bangladesh	1900–2014	663	3,330	1.43
United Kingdom	1870–2014	5,117	39,040	1.42

[a]Real GDP is measured in 2014 dollars.

Economic Growth around the World

- Annual growth rates of real GDP for China, Germany, Japan, the UK and the US



Economic Growth around the World

the UK

- Income per person was **$39,040**.
- **4** out of 1,000 children die before reaching age 5.
- **Almost the entire** population has access to modern sanitation facilities, such as a bathroom and sewer system, to safely remove human waste.
- Educational attainment is high: Among individuals of college age, **60 percent** are enrolled in higher education.

Economic Growth around the World

Mexico

- Income per person was **$16,640**.
- About **13** out of 1,000 children die before age 5.
- About **85 percent** have access to modern sanitation.
- Among those of college age, **30 percent** are enrolled.

Economic Growth around the World

Mali

- Income per person was only **$1,510**.
- **115** out of 1,000 children die before age 5.
- **25 percent** of the population has access to modern sanitation.
- Among those of college age, only **7 percent** are enrolled.

Productivity

- Productivity
  - Quantity of goods and services
  - Produced from each unit of labor input
- Why productivity is so important
  - Key determinant of living standards
  - Growth in productivity is the key determinant of growth in living standards
  - An economy's income is the economy's output

Productivity

- Determinants of productivity
  - Physical capital per worker
  - Human capital per worker
  - Natural resources per worker
  - Technological knowledge

Productivity

- Physical capital
  - Stock of equipment and structures
  - Used to produce goods and services
- Human capital
  - Knowledge and skills that workers acquire through education, training, and experience

Productivity

- Natural resources
  - Inputs into the production of goods and services
  - Provided by nature, such as land, rivers, and mineral deposits
- Technological knowledge
  - Society's understanding of the best ways to produce goods and services

Productivity

- Production function

$$Y = AF(L, K, H, N)$$

  - where Y is output
  - AF is technology
  - L is labor
  - K is capital
  - H is human capital
  - N is natural resources

Economic Growth and Public Policy

Saving and Investment

- Raise future productivity
  - Invest more current resources in the production of capital
  - Trade-off
  - Devote fewer resources to produce goods and services for current consumption
  - Consider higher savings rate;
    - Fewer resources – used to make consumption goods
    - More resources – to make capital goods
    - Capital stock increases
    - Rising productivity
    - More rapid growth in GDP

Economic Growth and Public Policy

Diminishing returns

- Benefit from an extra unit of an input
- Declines as the quantity of the input increases
- In the long run, higher savings rate
  - Higher level of productivity
  - Higher level of income
  - Not higher growth in productivity or income

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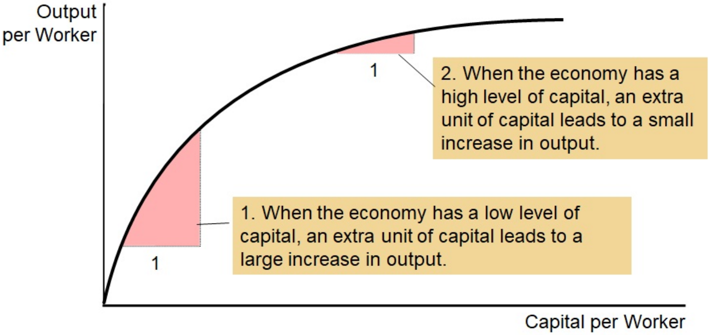

Economic Growth and Public Policy

Diminishing returns

- Catch-up effect
  - Countries that start off poor
  - Tend to grow more rapidly than countries that start off rich

- Poor countries
  - Low productivity
  - Even small amounts of capital investment
  - Increase workers' productivity substantially

Economic Growth and Public Policy

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Economic Growth and Public Policy

Investment from abroad

- Another way for a country to invest in new capital
- Foreign direct investment
  - Capital investment that is owned and operated by a foreign entity
- Foreign portfolio investment
  - Investment financed with foreign money but operated by domestic residents

Economic Growth and Public Policy

Investment from abroad

- Benefits from investment
  - Some flow back to the foreign capital owners
  - Increase the economy's stock of capital
  - Higher productivity
  - Higher wages
  - State-of-the-art technologies

Economic Growth and Public Policy

Education

- Investment in human capital
  - Gap between wages of educated and uneducated workers
  - Opportunity cost: wages forgone
  - Conveys positive externalities
  - Public education – large subsidies to human-capital investment

Economic Growth and Public Policy

Health and Nutrition

- Healthier workers
  - More productive
- Right investments in the health of the population
  - Increase productivity
  - Raise living standards
- Vicious circle in poor countries
  - Poor countries are poor: Because their populations are not healthy
  - Populations are not healthy: Because they are poor and cannot afford better healthcare and nutrition
- Virtuous circle
  - Policies that lead to more rapid economic growth
  - Would naturally improve health outcomes, which in turn would further promote economic growth

Economic Growth and Public Policy

Property Rights, Political Stability

- To foster economic growth
  - Protect property rights
  - Ability of people to exercise authority over the resources they own
  - Courts – enforce property rights
- Promote political stability
- Property rights
  - Prerequisite for the price system to work

Economic Growth and Public Policy

Free Trade

- Inward-oriented policies
  - Avoid interaction with the rest of the world
  - Infant-industry argument: Tariffs, and other trade restrictions
  - Adverse effect on economic growth
- Outward-oriented policies
  - Integrate into the world economy
  - International trade in goods and services
  - Economic growth

Economic Growth and Public Policy

Research and Development

- Government – encourages research and development
  - Farming methods
  - Aerospace research (Air Force; NASA)
  - Research grants
  - Tax breaks
  - Patent system

Economic Growth and Public Policy

Population Growth

- Large population
  - More workers to produce goods and services
  - Larger total output of goods and services
  - More consumers
- Stretching natural resources
  - Malthus: an ever-increasing population
  - Strain society's ability to provide for itself
  - Mankind – doomed to forever live in poverty

Economic Growth and Public Policy

Population Growth

- Diluting the capital stock
  - High population growth
  - Spread the capital stock more thinly
  - Lower productivity per worker
  - Lower GDP per worker
- Reducing the rate of population growth
  - Government regulation
  - Increased awareness of birth control
  - Equal opportunities for women

- Promoting technological progress
  - World population growth
  - Engine for technological progress and economic prosperity
  - More people = More scientists, more inventors, more engineers